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Floquet Theory applied to Lindbladian Open Quantum Systems

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AGENCIA ESTATAL DE INVESTIGACIÓN
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Motivation

	Closed	Open Lindbladian
State		
Evolution		
Floquet		

CPTP ≡ Completely Positive Trace Preserving Map

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Is the Floquet Generator a Lindbladian?

CPTP

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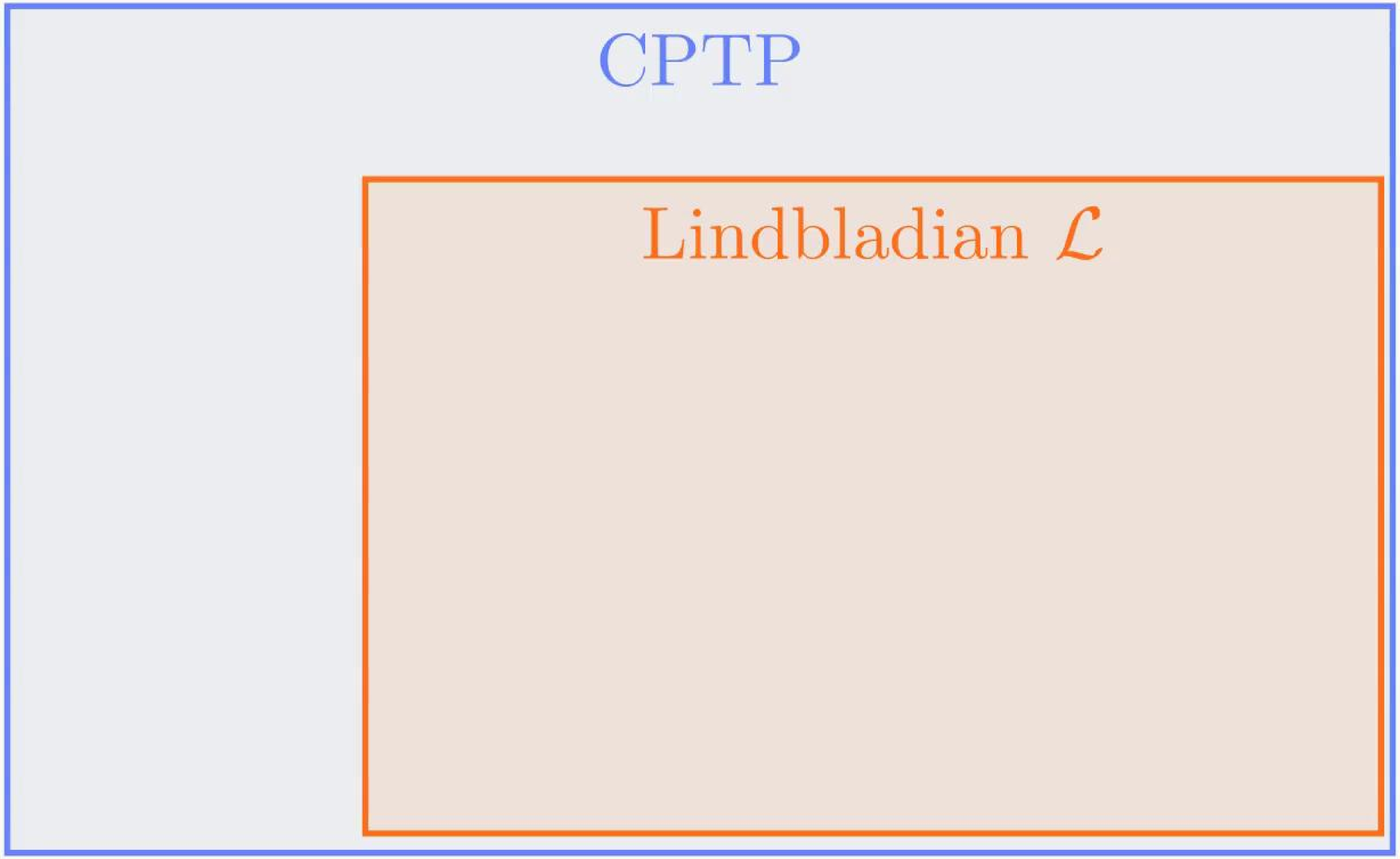

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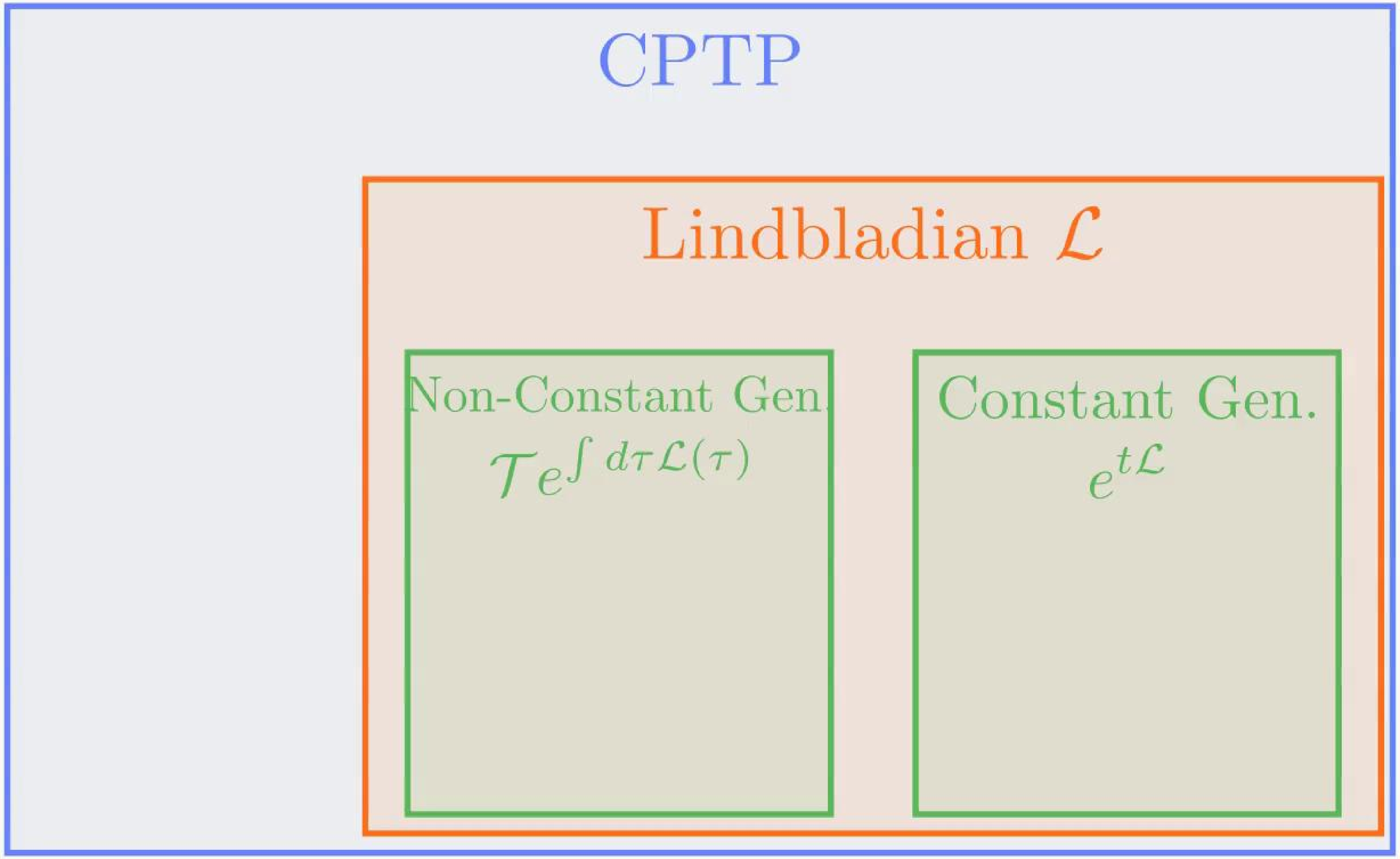

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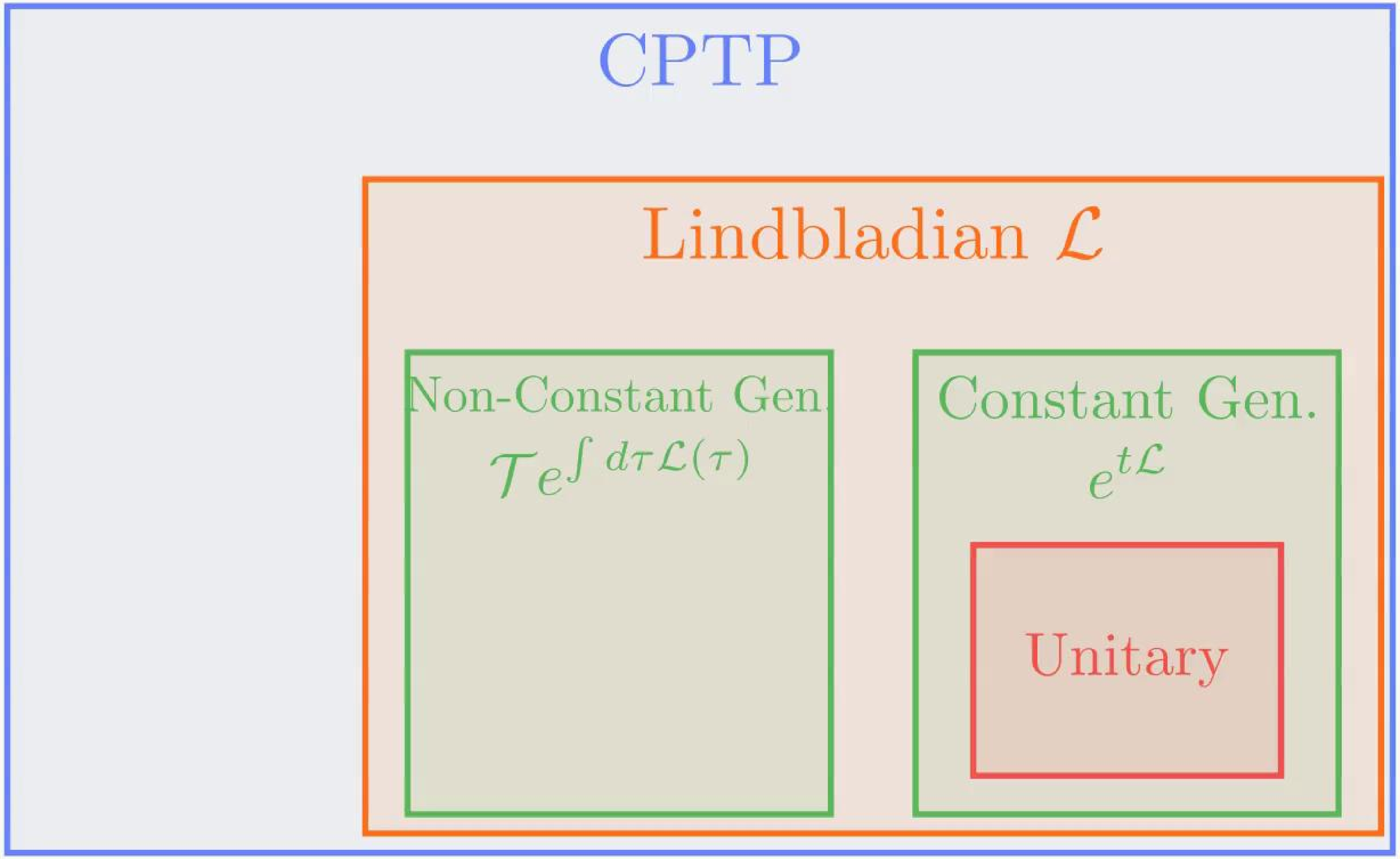

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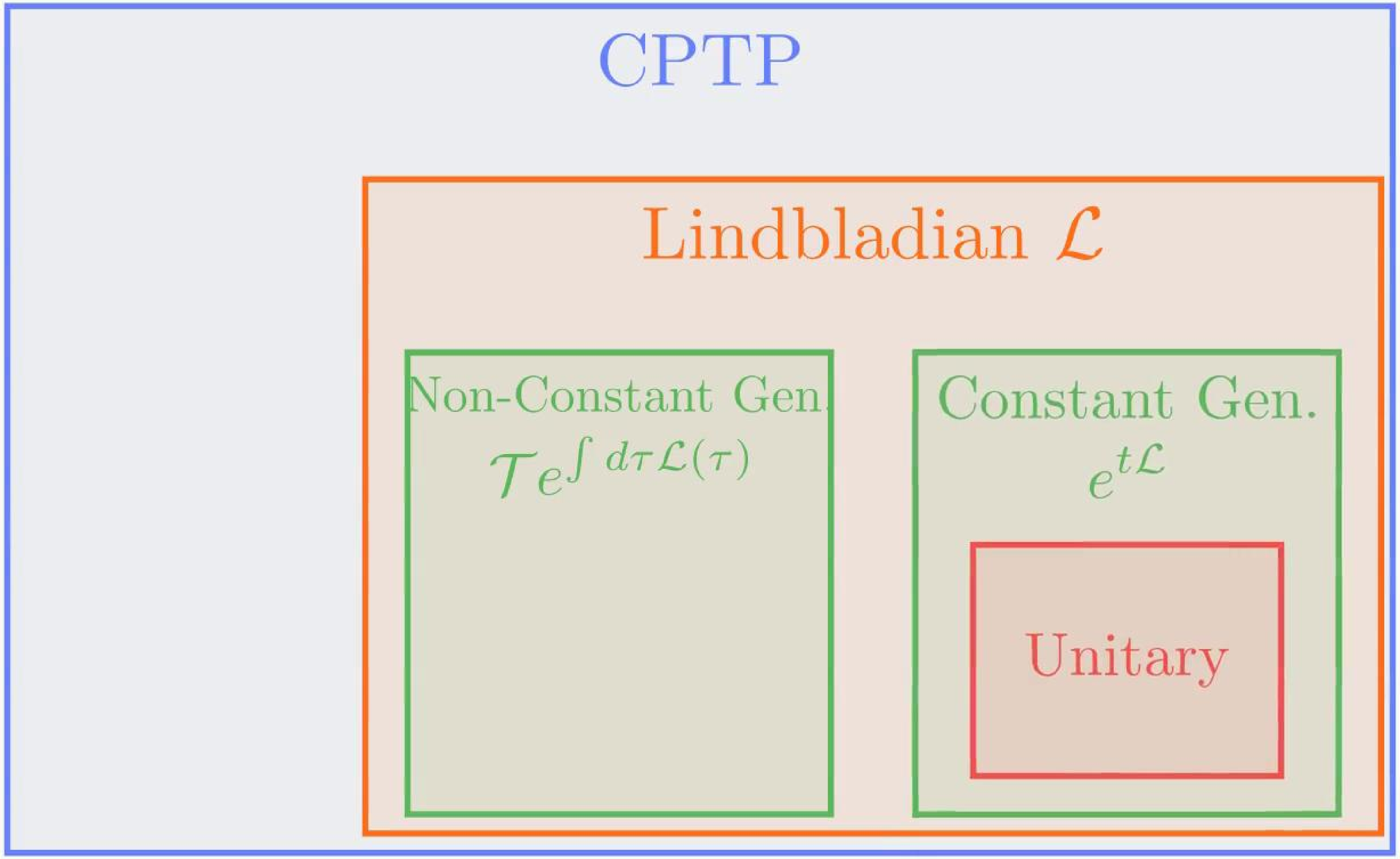

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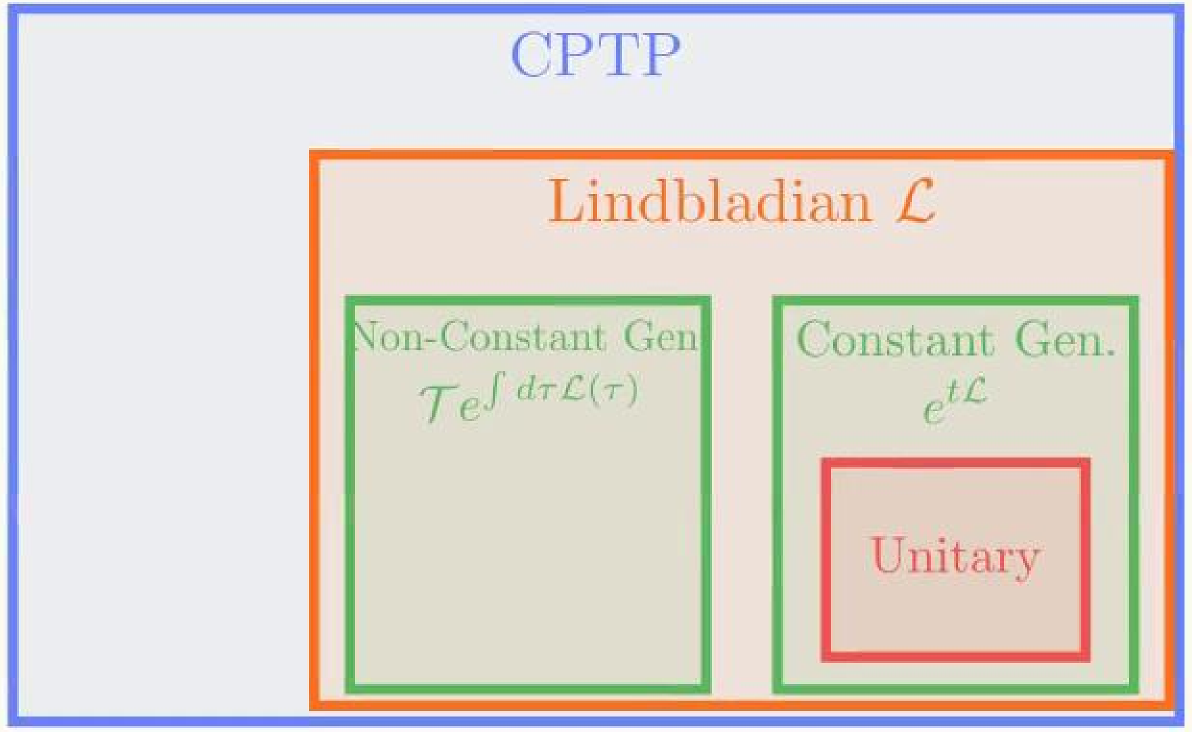

[1] Wolf, M., Eisert, J., Cubitt, T., & Cirac, J. (2008). *Assessing Non-Markovian Quantum Dynamics.* Phys. Rev. Lett., 101, 150402.

[2] Schnell, A., Eckardt, A., & Denisov, S. (2020). *Is there a Floquet Lindbladian?* Phys. Rev. B, 101, 100301.

Necessary and sufficient conditions given by [1]

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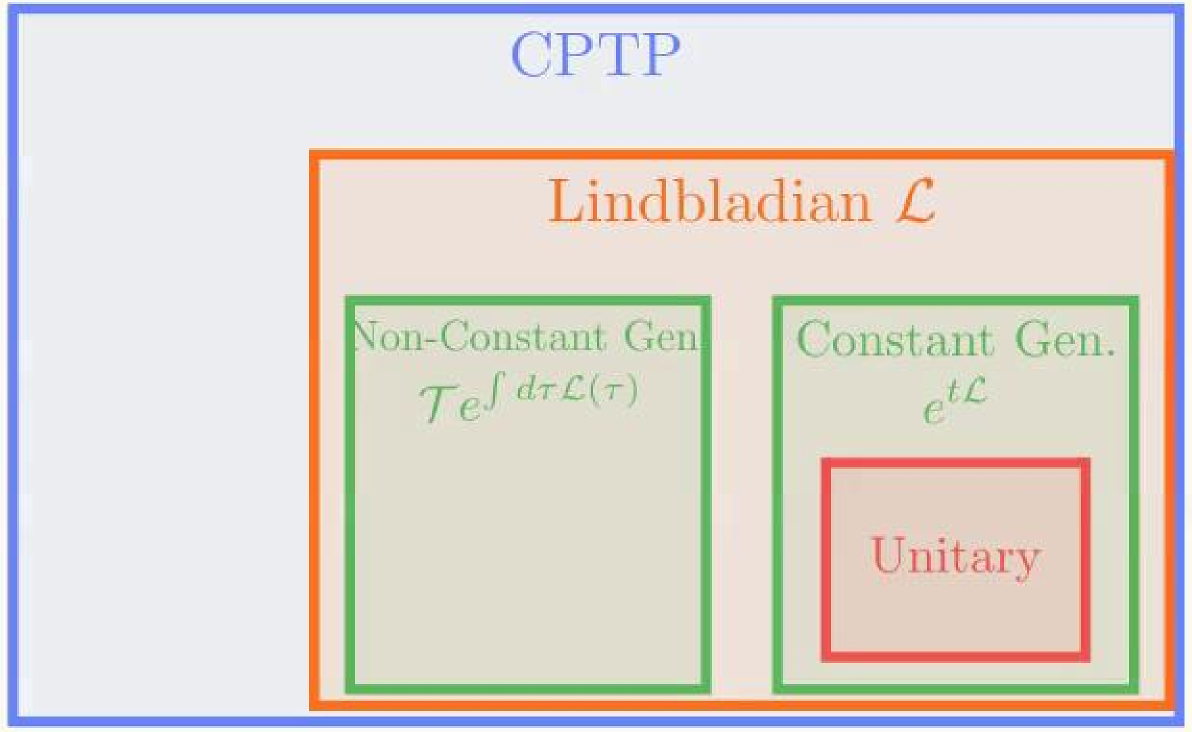

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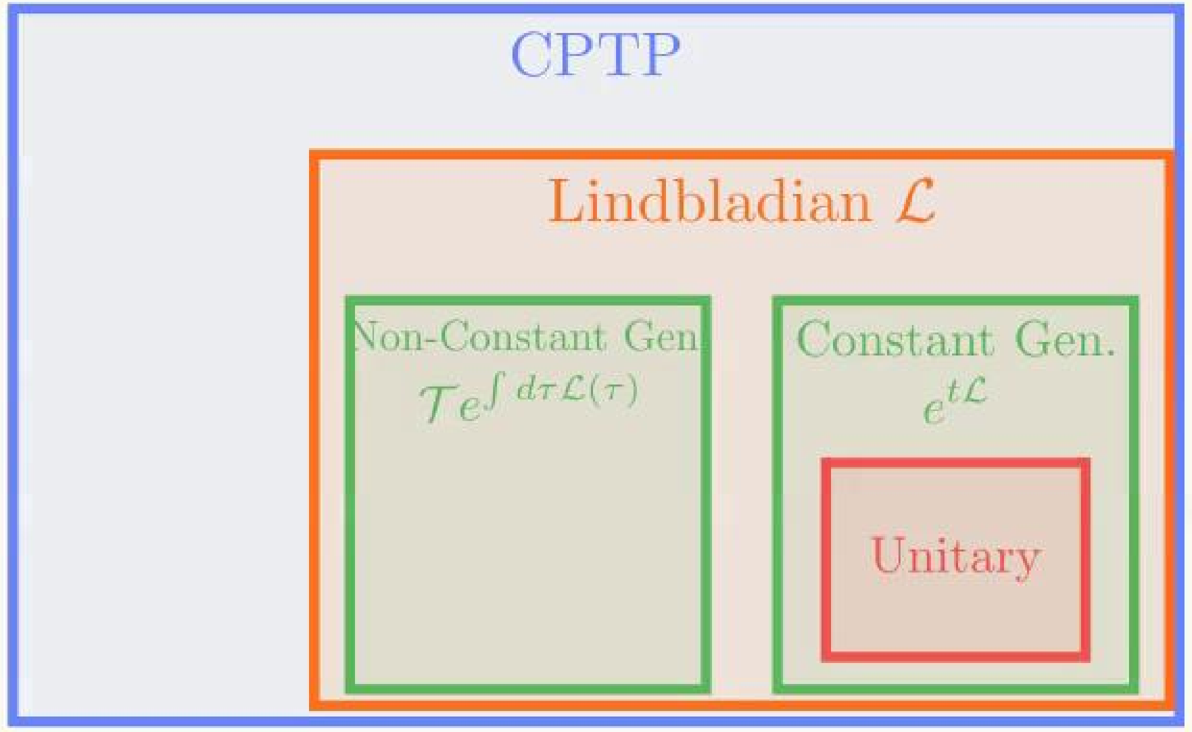

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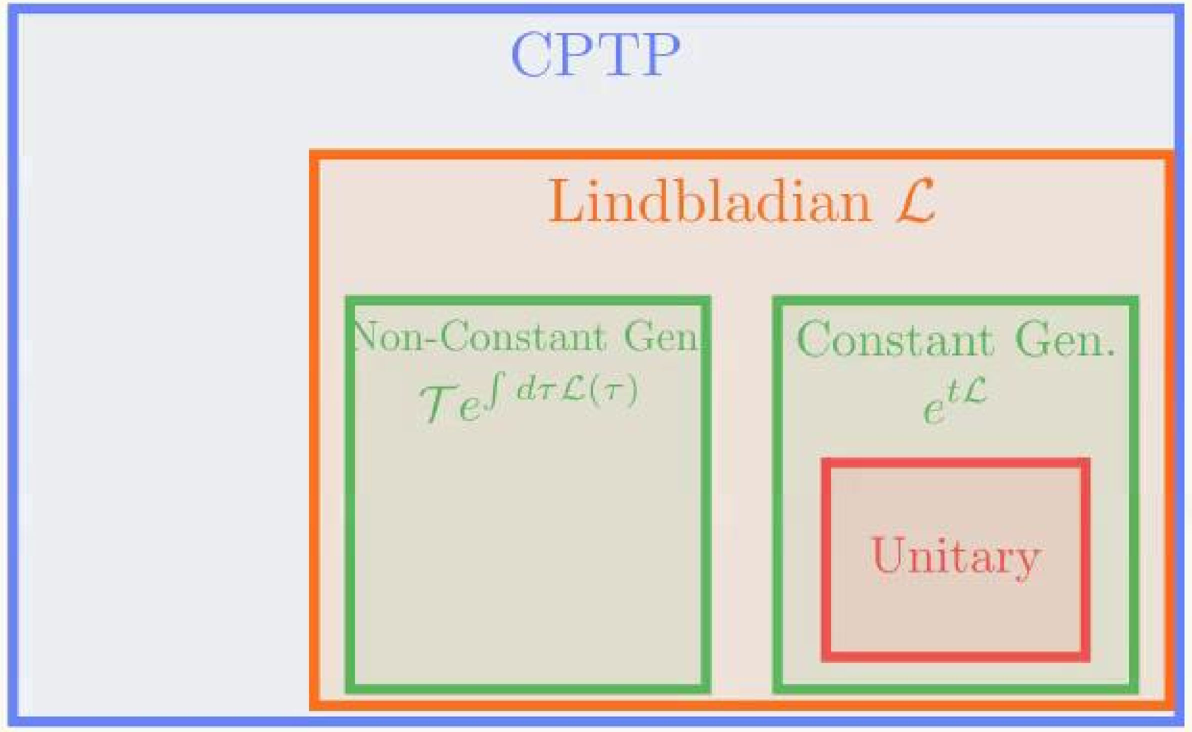

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Only valid for diagonalizable cases → Let's generalize it. Symmetries

Thank you for your attention!
Any questions?

Floquet Theorem

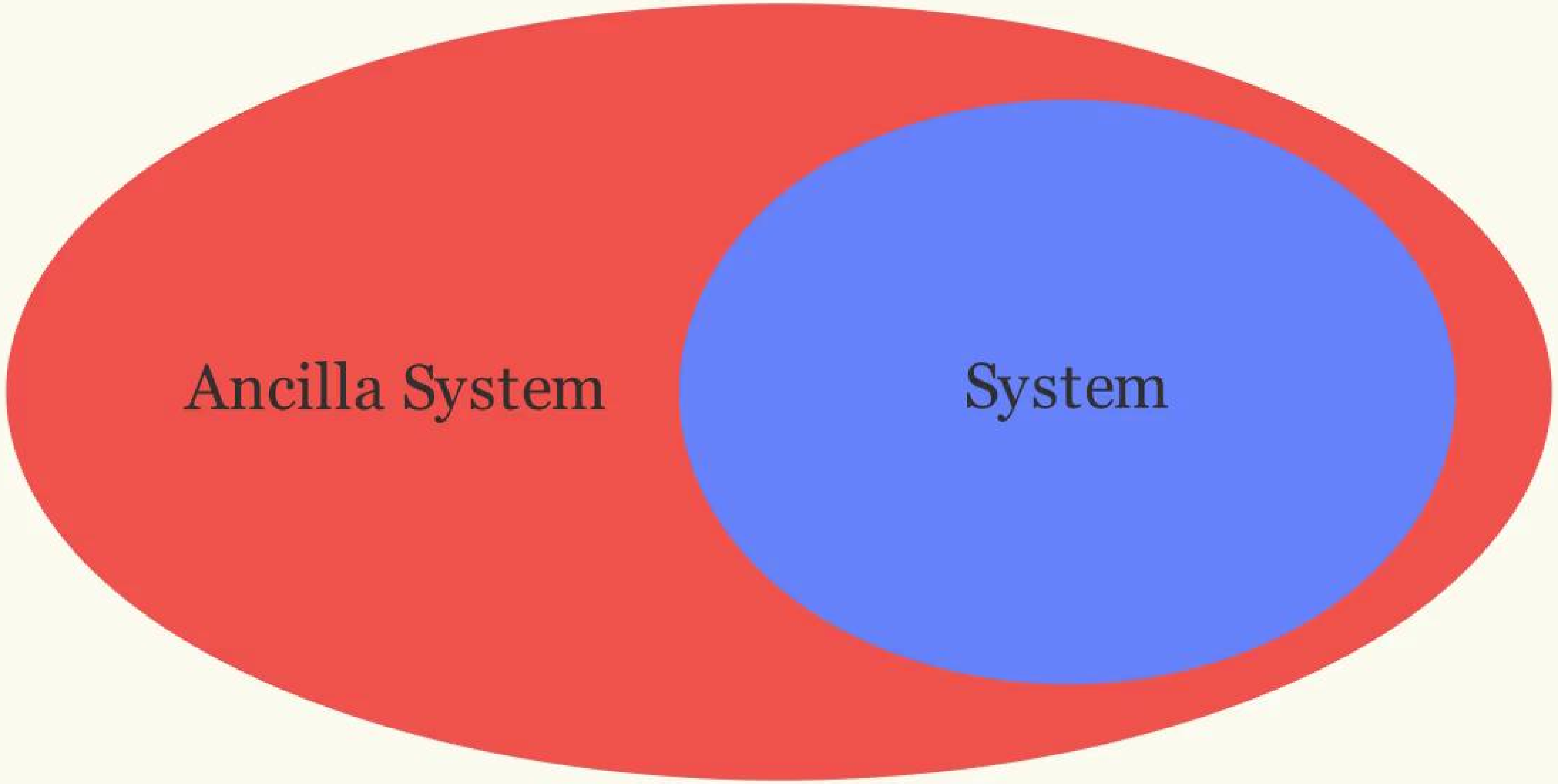
Consider the system of linear diff. equations for $M(t) \in \mathcal{M}_n(\mathbb{C})$

$$\frac{d}{dt}M(t) = C(t)M(t); \quad M(0) = I_n; \qquad C(t) \in \mathcal{M}_n(\mathbb{C})$$

If $M(t) = M(t+T)$ for some fixed $T \in \mathbb{R}$, then:

$$M(t) = P(t)e^{tG} \text{ with } P(t) = P(t+T)$$

Completely Positive Maps



$\mathcal{E} \otimes \mathcal{I}_{\mathrm{n}}$ must map density matrices to density matrices